Docker Setup

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Docker Background

Docker is a **virtualization** engine that allows us to run tools in *containers*

This lets us safely run a standardized operating system on our machines without having to replace our operating system

This prevents “works on my machine” problems and ensures your code runs the same way in the grading environment

Terminology

- **Virtualization**: Running a program or operating system in a way that it *thinks* it's running on real hardware
  - Note: We'll see other types of virtualization in class (memory, storage, etc.)
- **Image**: A template containing an operating system and pre-installed tools (like `samogden/cst334`)
- **Container**: A running instance of an image that you can interact with
- **Volume/Mount**: Connecting a folder from your computer to the container so files are shared between them
- **Docker Daemon**: A background service that manages containers (must be running to use Docker)

What we use in class

We use a docker image named `samogden/cst334`

It contains all the tools we use and I use it for grading, so you can test in the same environment

Installing Docker

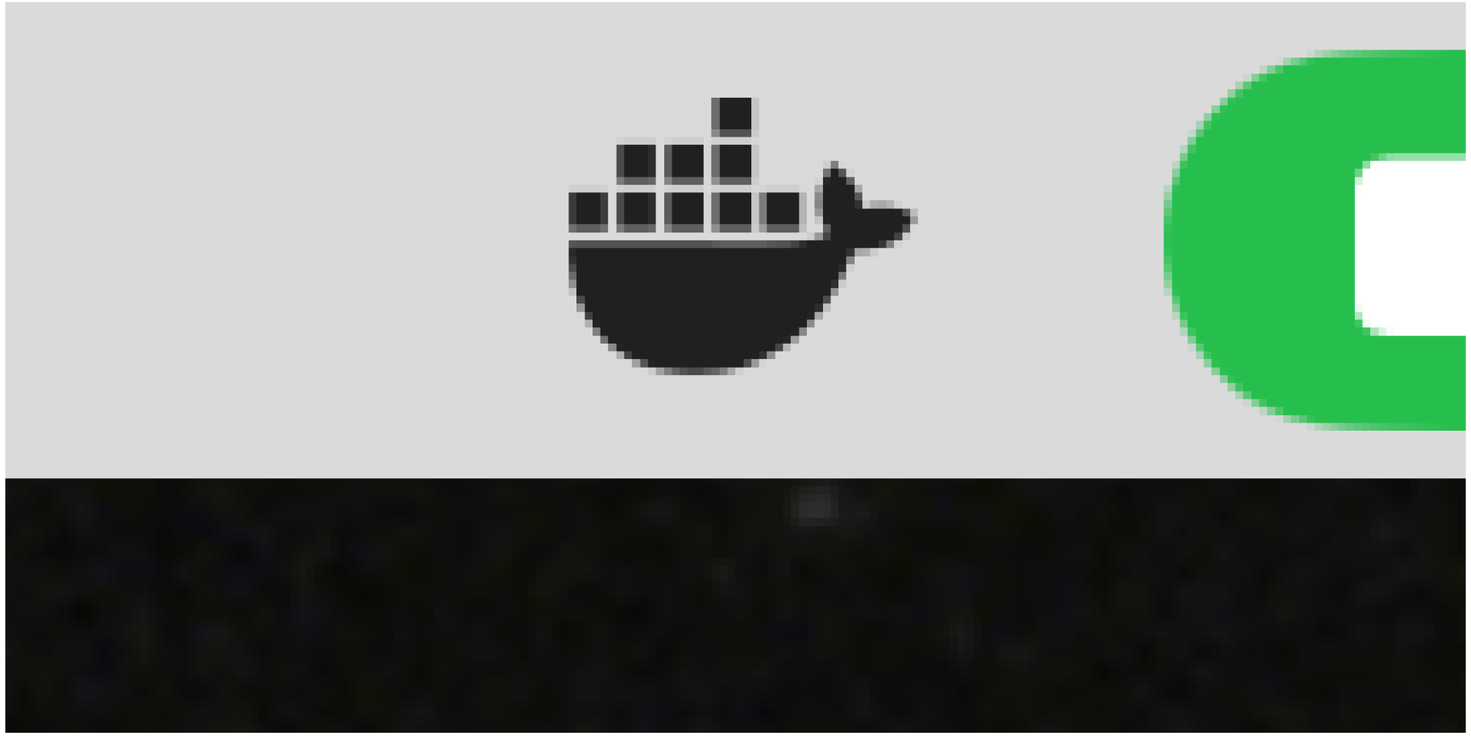
- Windows install instructions
- Mac install instructions
- Linux install instructions

Note: Before installing make sure you backup important data!

Starting Docker

To run docker you need generally need to open `Docker Desktop`

- You do **not** need to log in to Docker Desktop, even though they ask
- When you see the icon below in your taskbar or menu bar, Docker is running and ready to use



Testing your installation (part 1)

The best way to test your installation is to run `docker run hello-world` in your terminal[1]

Testing your installation (part 2)

```
$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
198f93fd5094: Pull complete
95ce02e4a4f1: Download complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (arm64v8)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
```

Note: The `$` at the very beginning of the above indicates that I was typing a command into terminal

Before you run docker for class

Before you run the docker command for class make sure you are not in your home directory!

You should open terminal and type in something like `cd Documents` to change to the documents folder.

Running docker in class

The command we use in class is:

```
1 docker run -it --rm -v ${PWD}:/tmp/hw samogden/cst334
```

This command has a few parts we'll go over in the next slide

[^]: **Note for Windows users**: `${PWD}` works in PowerShell. If you have issues, try `${pwd}` (lowercase).

Running docker in class (core command)

- `docker` : this invokes the docker program
- `run` : this tells the docker program you want to start a container from an image
- `samogden/cst334` : the name of the docker image we are using in class

Running docker in class (flags)

- `-it` : run this container interactively (i.e. so we can type commands into it)
- `--rm` : tells docker to delete the container after we are no longer interacting with it
- `-v ${PWD}:/tmp/hw` : this tells docker to make the current directory available inside docker at the path `/tmp/hw`
  - `${PWD}` will be evaluated by the terminal to point to the current directory.
  - It stands for “print working directory”

Running Docker in class

What this looks like

```
$ docker run -it --rm -v ${PWD}:/tmp/hw samogden/cst334
[DOCKER] /tmp/ $
```

Note: You might see more between these two lines, particularly notes about downloading layers, which are due to docker having to download the image

Navigating inside Docker

Once inside Docker, navigate to your mounted directory:

```
[DOCKER] /tmp/ $ cd /tmp/hw
[DOCKER] /tmp/hw $ ls
labs  programming-assignments  README.md
[DOCKER] /tmp/hw $ cd labs/1-debugging_with_gdb
[DOCKER] /tmp/hw/labs/1-debugging_with_gdb $ make
```

All your files from the host machine are available at `/tmp/hw`

Updating the docker image

Every so often I push updates to docker images, usually for Quality of Life (QoL) additions.

You can update your docker image with:

```
1 docker pull samogden/cst334:latest
```

Exiting Docker

To exit docker you can either type in `exit`, press `control-d` or simply close the terminal.

Note that when you exit docker the container itself is lost, but your data should still in the folder you started in (thanks to the `-v` flag discussed earlier).

Complete Workflow Example

Putting it all together:

```
# On your host machine - navigate to your course directory
$ cd ~/Documents/CST334/labs/1-debugging_with_gdb

# Start Docker with current directory mounted
$ docker run -it --rm -v ${PWD}:/tmp/hw samogden/cst334

# Inside Docker - navigate to mounted directory
[DOCKER] /tmp/ $ cd /tmp/hw

# Build and test your code
[DOCKER] /tmp/hw $ make
[DOCKER] /tmp/hw $ make grade

# Exit when done
[DOCKER] /tmp/hw $ exit
$
```

Troubleshooting

Docker daemon is not running - Make sure Docker Desktop is open and running (look for the whale icon)

Permission denied (Linux) - You may need to add your user to the docker group: `sudo usermod -aG docker $USER` - Log out and back in for changes to take effect

Cannot connect to Docker daemon (Mac) - Restart Docker Desktop from Applications

Files not showing up in `/tmp/hw` - Make sure you're running the docker command from the correct directory - Check that you included the `-v ${PWD}:/tmp/hw` flag